Note

1. If a circle of radius r units divided into n equal sectors, then the

Length of the arc of each of the sectors $= \frac{1}{n} \times 2\pi r \ units$ and

Area of each of the sectors $= \frac{1}{n} \times \pi r^2 \ sq.units$

2. Also, the area of the sector is derived as $= \frac{\theta^\circ}{360^\circ} \times \pi r^2$

$$= \frac{1}{2}\left(\frac{\theta^\circ}{360^\circ} \times 2\pi r\right) \times r$$

$$= \left(\frac{1}{2} \times l\right) \times r = \frac{lr}{2} \ sq.units$$

2.2.3 Perimeter of a sector

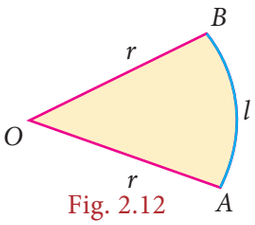


Fig. 2.12

We already know that the total length of the boundary of a closed part is its perimeter, Isn't it? What is the boundary of a sector? **Two radii** (OA and OB) and **an arc** $\left(\overset{\frown}{AB}\right)$.

So, the perimeter of a sector = length of the arc + length of two radii

$$P = l + 2r \ units$$

∴ **Perimeter of a sector,** $P = l + 2r \ units$

DO YOU KNOW?

Tamil people always have a prominent role in the history of Mathematics. They had recorded in the form of a song on how to find the area of a circle, a few thousand years before itself, in the book titled 'Kanakkathikaram'. (கணக்கதிகாரம்). The song is as follows:

"வட்டத்தரை கொண்டு விட்டத்தரை தாக்கச்
சட்டெனத் தோன்றும் குழி "

Meaning:

'வட்டத்தரை' represents half of the circumference and **'விட்டத்தரை'** represents half of the diameter which is the radius and **'குழி'** represents the area. In this song, the Tamil people had recorded that, if half the circumference is multiplied by half the diameter, the area of the circle can be calculated.

Area of the circle = **வட்டத்தரை** × **விட்டத்தரை**

$$= \frac{1}{2} \text{ of circumference} \times \frac{1}{2} \text{ of diameter} = \left(\frac{1}{2} \times 2\pi r\right) \times r$$

∴ Area of the circle, $A = \pi r^2 \ sq.units$

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